

LECTURE 2: Conditioning and Bayes' rule

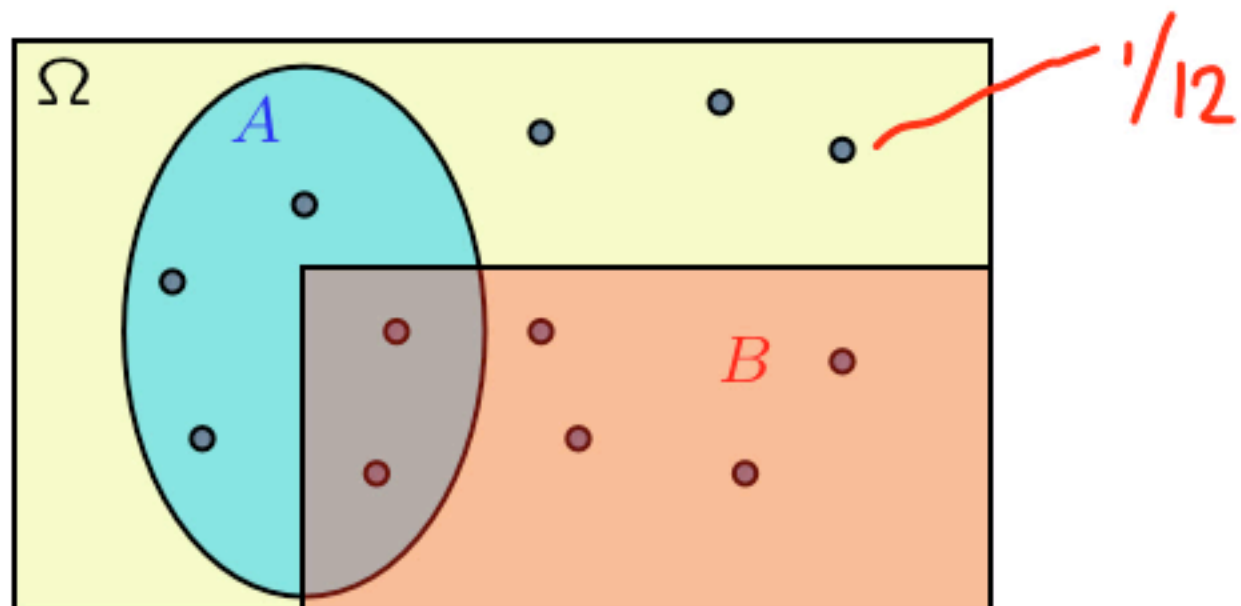
- Conditional probability
- Three **important** tools:
 - Multiplication rule
 - Total probability theorem
 - Bayes' rule ($\longrightarrow$ inference)

<https://betterexplained.com/articles/an-intuitive-and-short-explanation-of-bayes-theorem/>

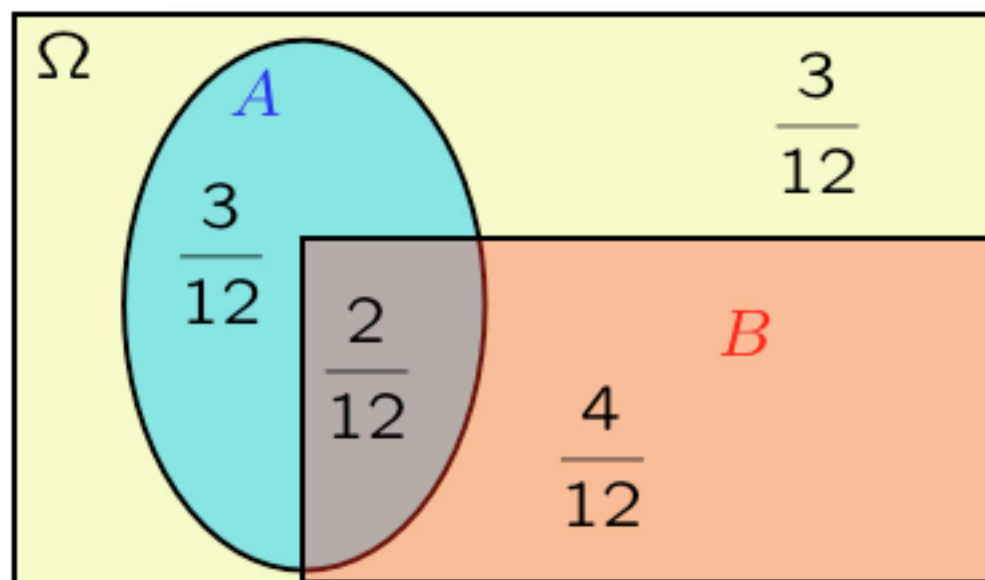
The idea of conditioning

Use new information to revise a model

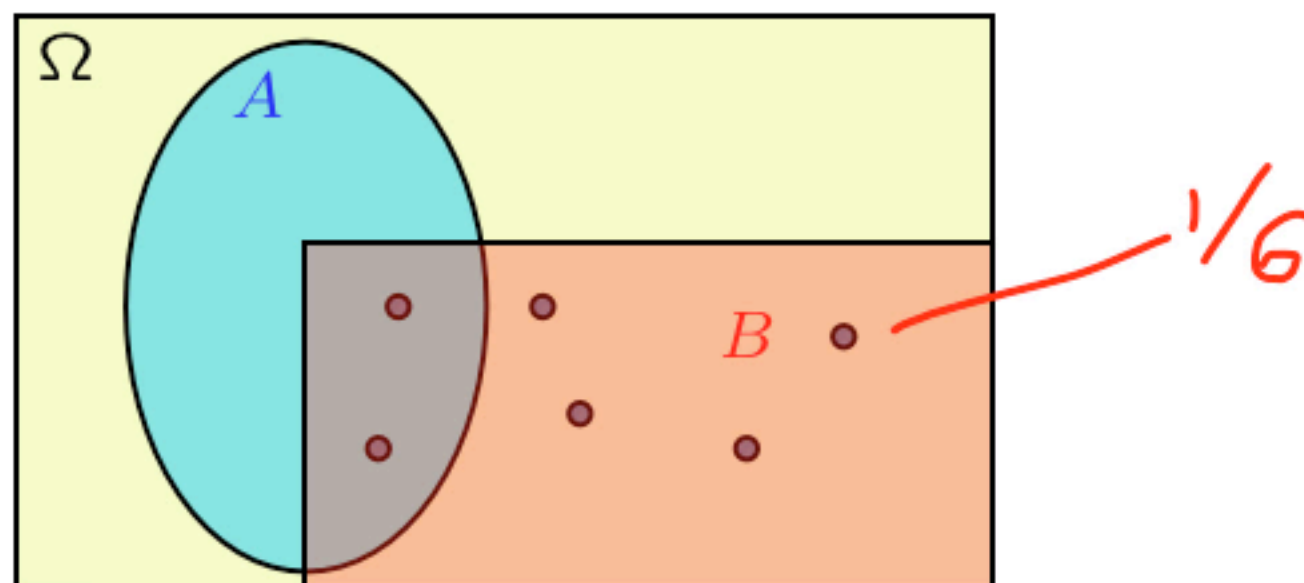
Assume 12 equally likely outcomes



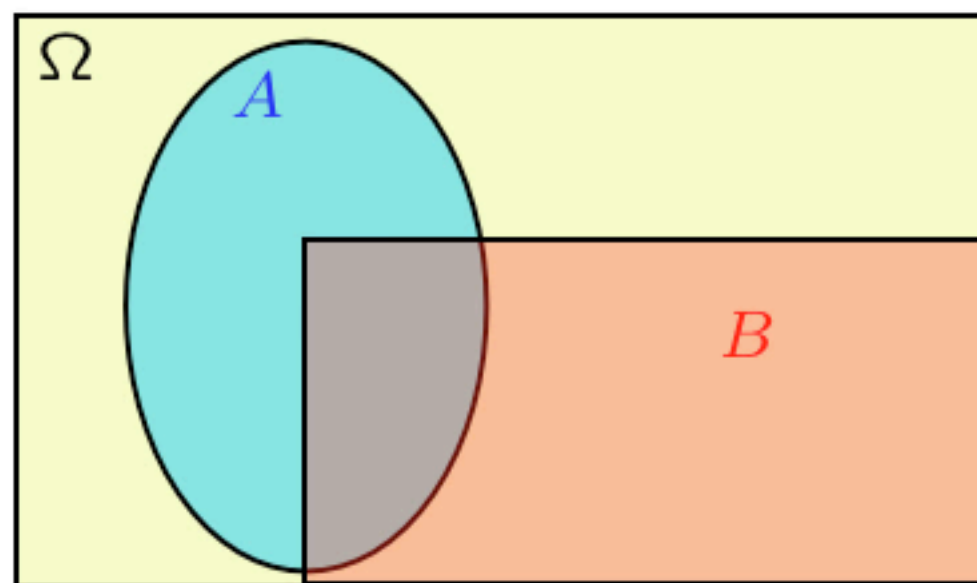
$$P(A) = \frac{5}{12} \quad P(B) = \frac{6}{12}$$



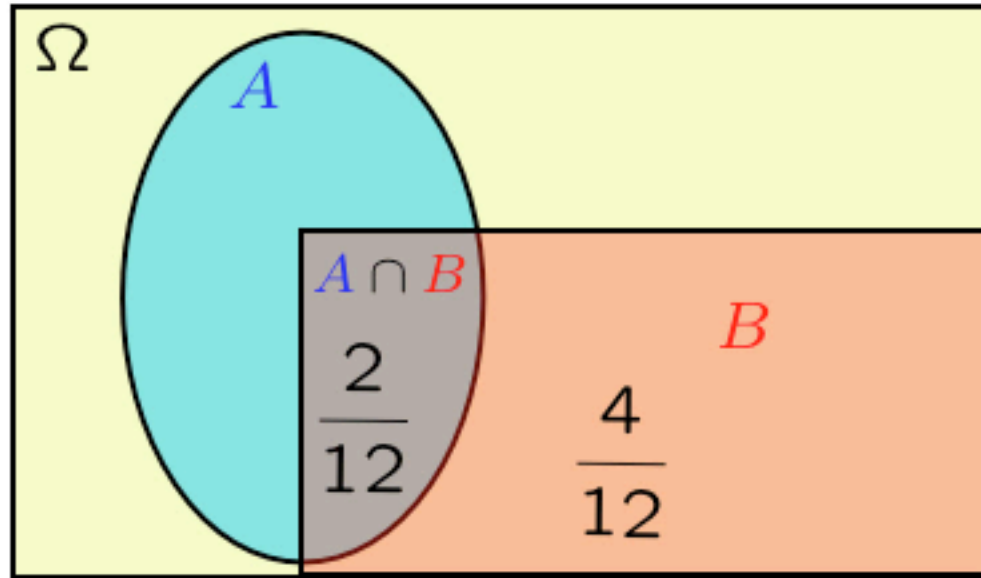
If told B occurred:



$$P(\underline{A} | \underline{B}) = \frac{2}{6} = \frac{1}{3} \quad P(\underline{B} | \underline{B}) = 1$$



Definition of conditional probability



- $P(A|B)$ = “probability of A , given that B occurred”

Def. $\rightarrow$

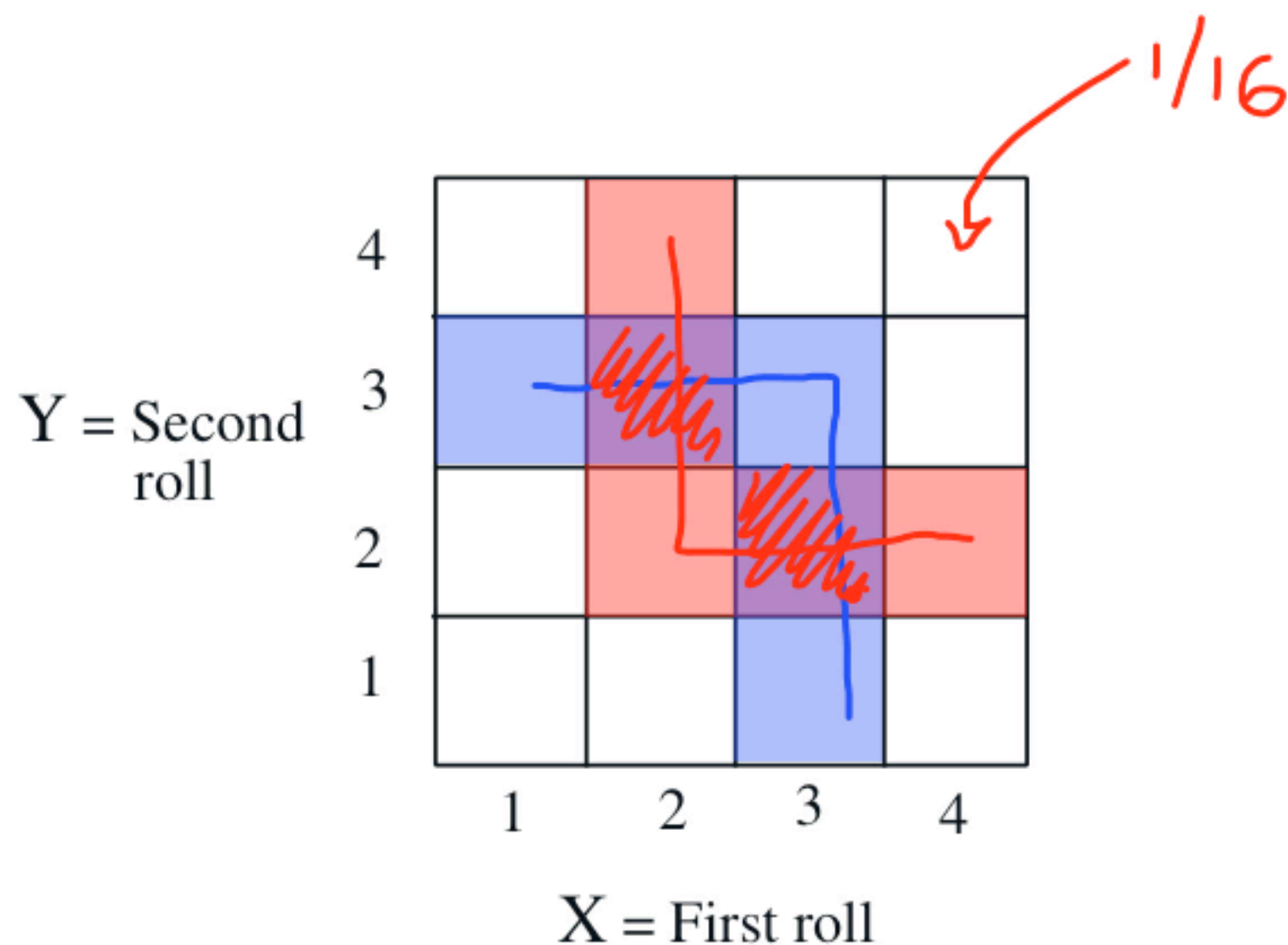
$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)} \leftarrow$$

defined only when $P(B) > 0$

$$= \frac{2/12}{4/12} = \frac{1}{2}$$

B é o novo espaço amostral

Example: two rolls of a 4-sided die



- Let B be the event: $\min(X, Y) = 2$

Let $M = \max(X, Y)$

$$P(M = 1 \mid B) = 0$$

$$P(\underline{M = 3} \mid B) = \frac{P(M=3 \text{ and } B)}{P(B)}$$

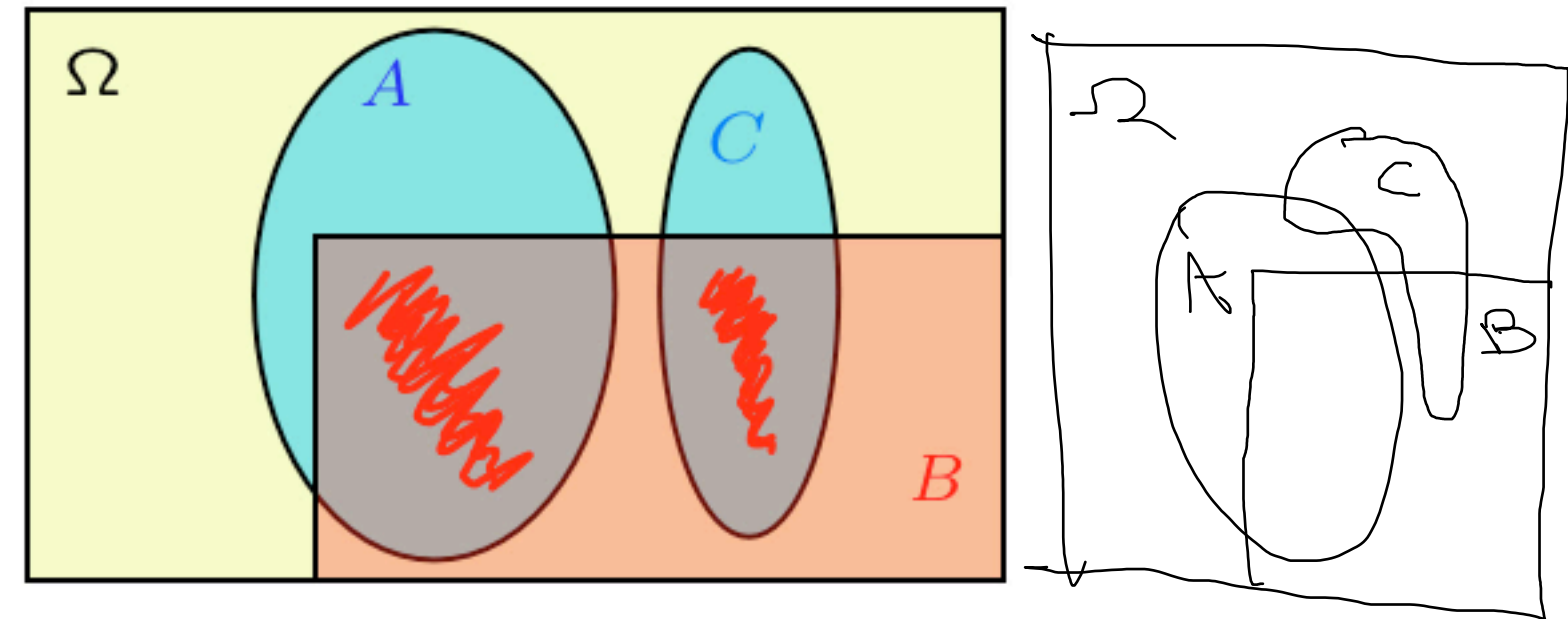
$$= \frac{2/16}{5/16} = \frac{2}{5}$$

Conditional probabilities share properties of ordinary probabilities

$$P(A | B) \geq 0 \quad \text{assuming } P(B) > 0$$

$$P(\Omega | B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

$$P(B | B) = \frac{P(B \cap B)}{P(B)} = 1$$



$$\text{If } A \cap C = \emptyset, \quad \text{then } P(\underbrace{A \cup C}_{\text{disjoint}} | \underbrace{B}_{\text{condition}}) = P(A | B) + P(C | B)$$

$$= \frac{P((A \cup C) \cap B)}{P(B)} = \frac{P(\underbrace{(A \cap B)}_{\text{disjoint}} \cup \underbrace{(C \cap B)}_{\text{disjoint}})}{P(B)} = \frac{P(A \cap B) + P(C \cap B)}{P(B)} =$$

$$= P(A | B) + P(C | B) \quad \text{also finite countable additivity}$$

Jaynes diz que toda probabilidade é condicional: ela depende do modelo usado, no mínimo: assumimos independência, distribuição normal, uma dada média, etc.

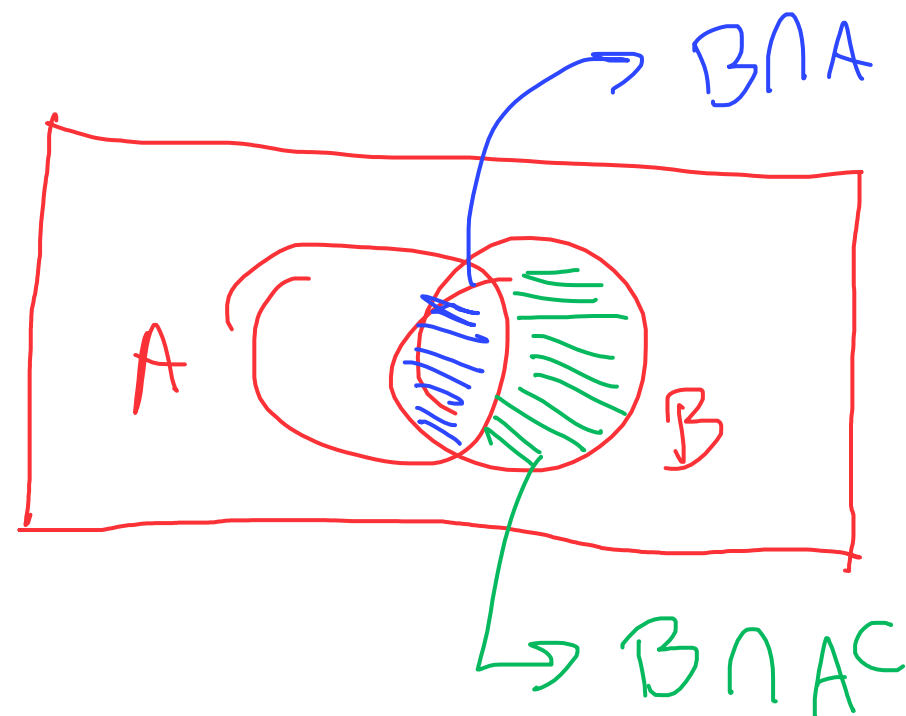
Confusão entre os condicionantes:

Prob(câncer|teste positivo) é muito diferente da Prob(teste positivo|câncer)

Prob(inglesa|rainha) é muito diferente da Prob (rainha|inglesa)

Prosecutor's fallacy: https://en.wikipedia.org/wiki/Prosecutor%27s_fallacy

Prob(evidências|inocente) é muito pequena, então não pode ser inocente. Mas o que se quer é Prob(inocente|evidências).



$$B = (B \cap A) \cup (B \cap A^c)$$
$$\Rightarrow P(B) = P(B \cap A) + P(B \cap A^c)$$

Models based on conditional probabilities

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad P(B | A) = \frac{P(A \cap B)}{P(A)}$$

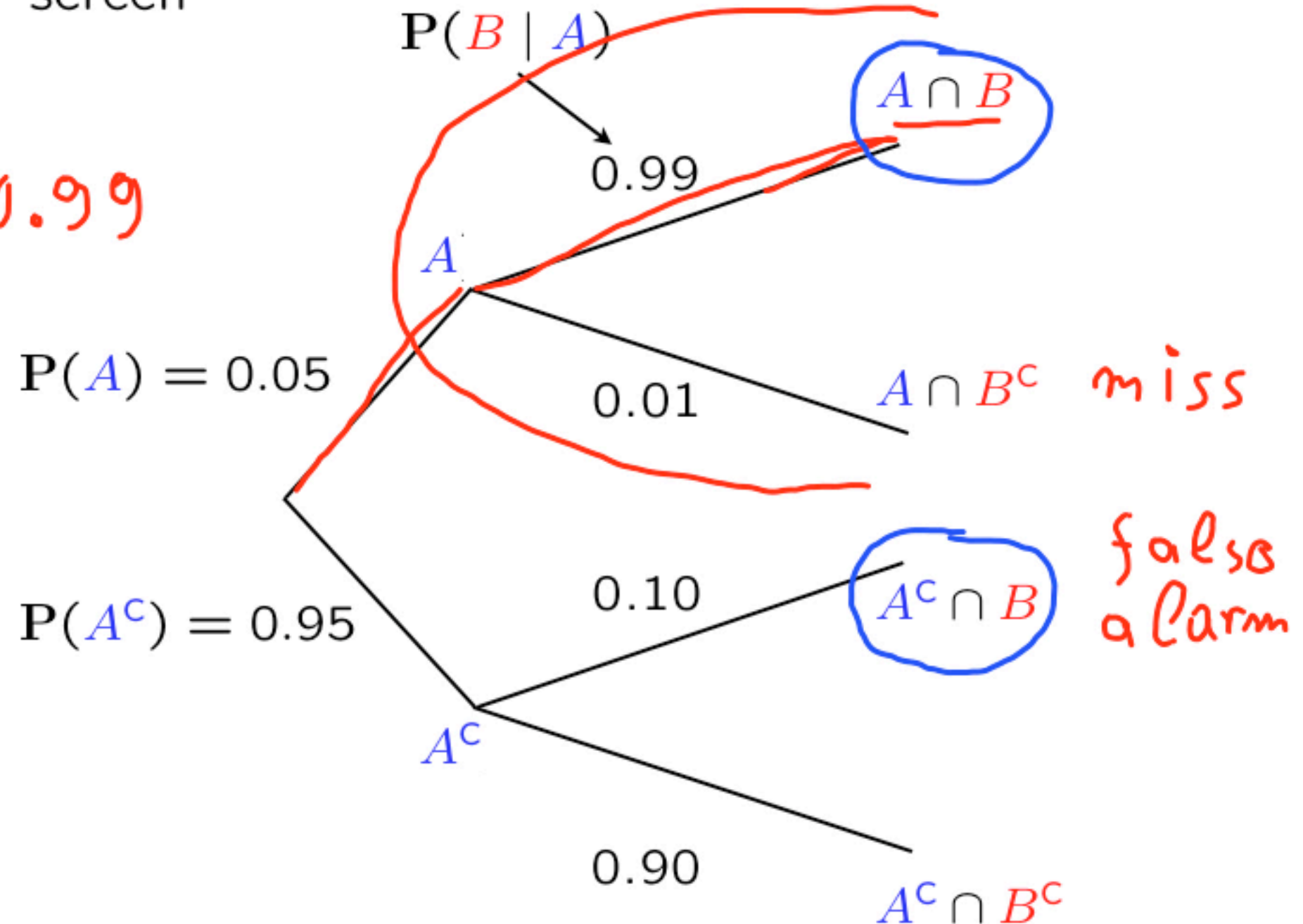
Event A : Airplane is flying above

Event B : Something registers on radar screen

- $P(A \cap B) = P(A) \cdot P(B|A) = 0.05 \cdot 0.99$

- $P(B) = \underbrace{0.05 \cdot 0.99}_{P(A \cap B)} + \underbrace{0.95 \cdot 0.1}_{P(A^c \cap B)} = 0.1445$

- $P(A | B) = \frac{0.05 \cdot 0.99}{0.1445} = 0.34$



The multiplication rule

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

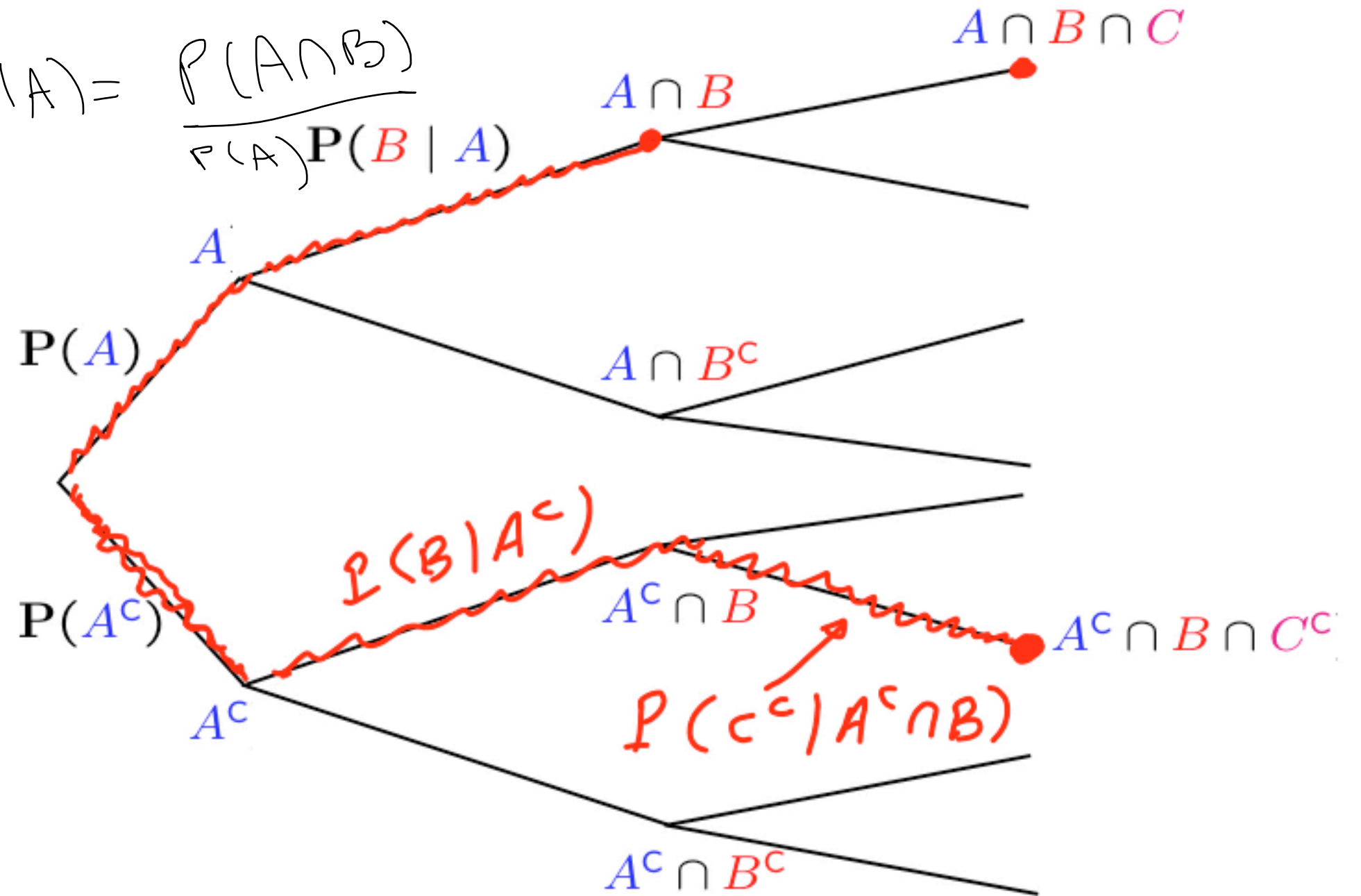
$$\begin{aligned} P(A \cap B) &= P(B) P(A | B) \\ &= P(A) P(B | A) \end{aligned}$$

$$P(\underbrace{A^c \cap B}_{\text{red}} \cap \underbrace{C^c}_{\text{red}}) =$$

$$= P(A^c \cap B) P(C^c | A^c \cap B)$$

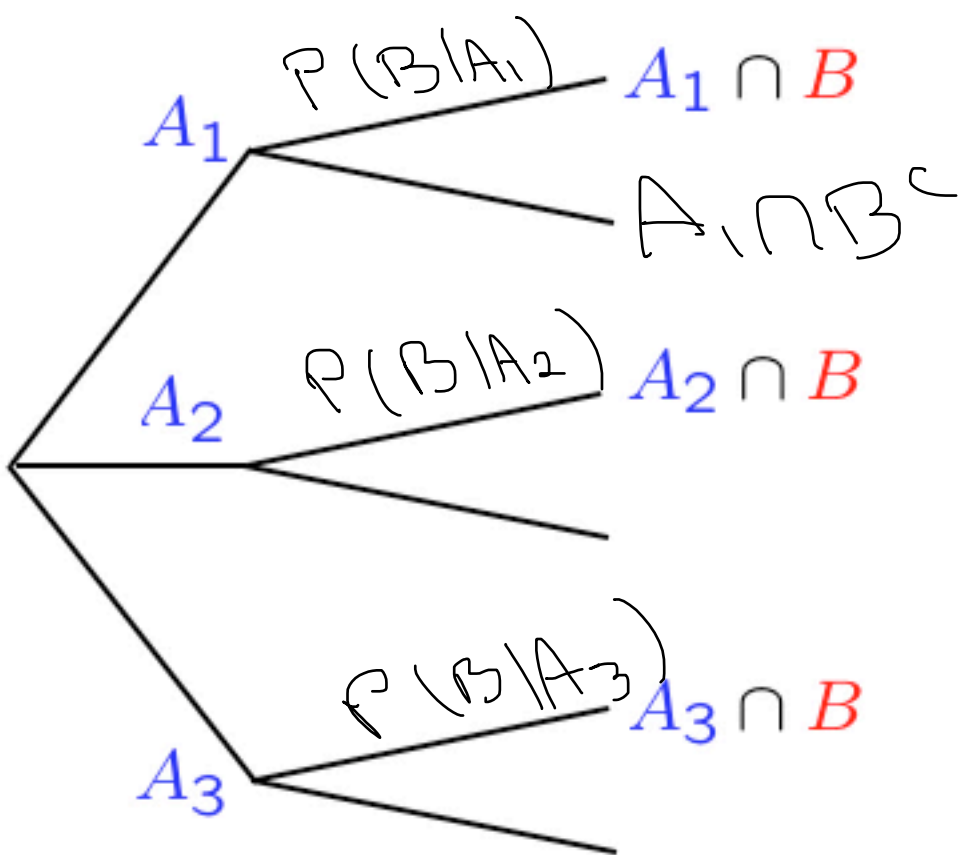
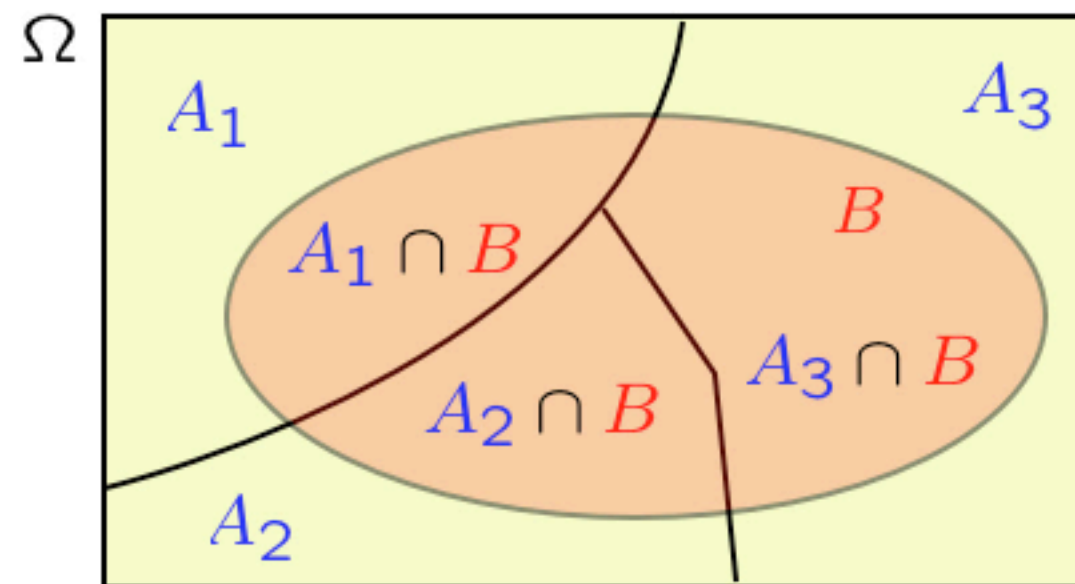
$$= P(A^c) \cdot P(B | A^c) P(C^c | A^c \cap B)$$

$$P(B | A) = \frac{P(A \cap B)}{P(A) P(B | A)}$$



$$\begin{aligned} &P(A_1 \cap A_2 \cap \dots \cap A_n) \\ &= P(A_1) \prod_{i=2}^n P(A_i | A_1 \cap \dots \cap A_{i-1}) \end{aligned}$$

Total probability theorem



- Partition of sample space into $A_1, A_2, A_3, \dots$
- Have $P(A_i)$, for every i
- Have $P(B | A_i)$, for every i

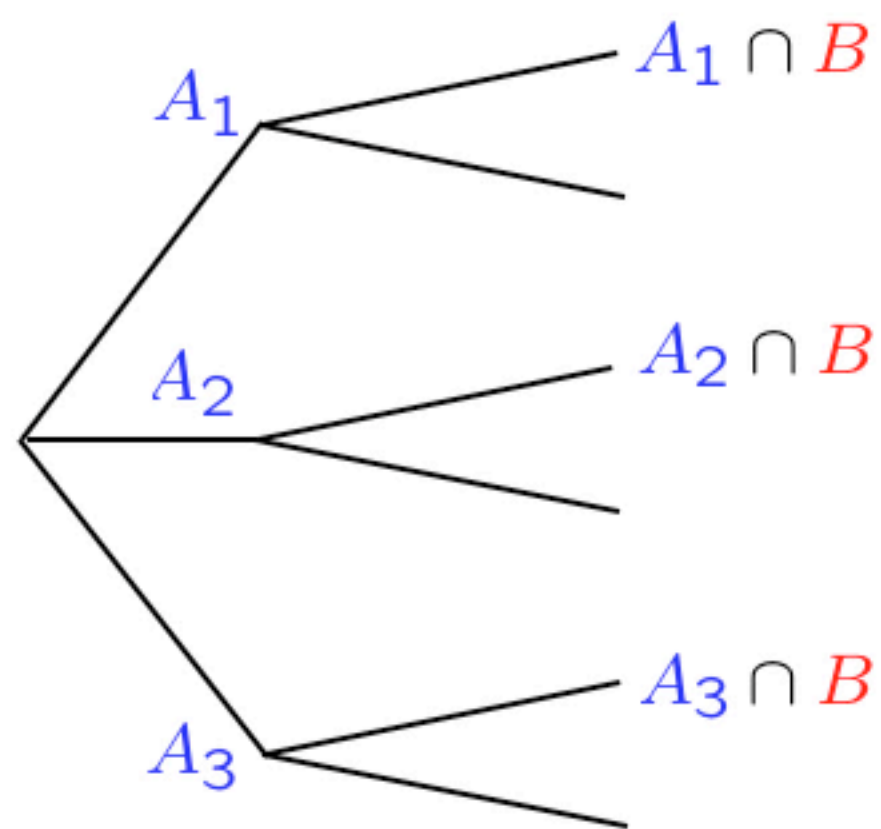
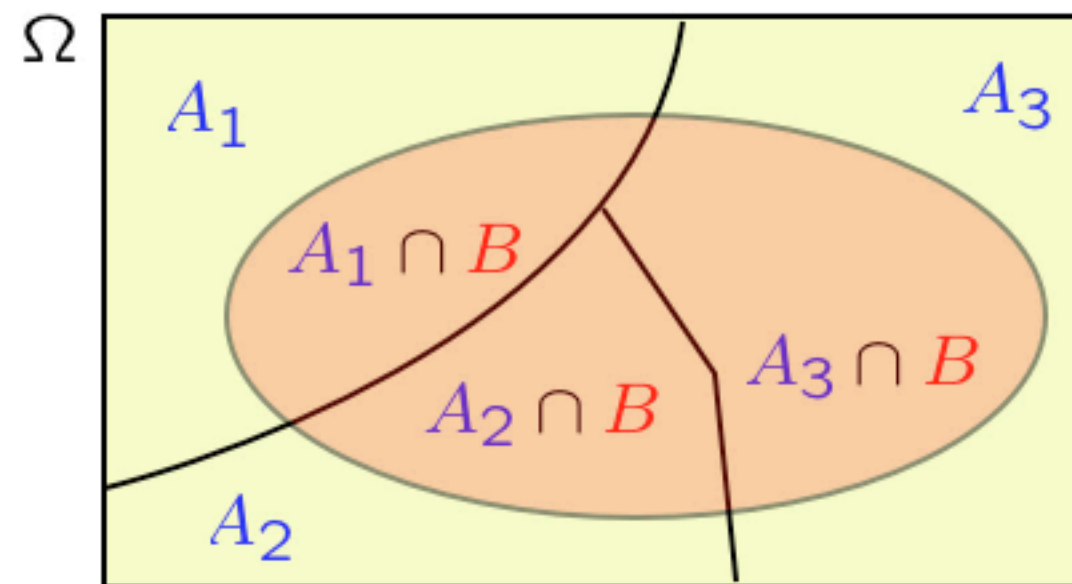
$$P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3) \\ = P(A_1)P(B|A_1) + \dots + \dots$$

$$\sum_i P(A_i) = 1$$

$$P(B) = \sum_i P(A_i) P(B | A_i)$$

weights
weighted average
of $P(B|A_i)$

Bayes' rule



- Partition of sample space into A_1, A_2, A_3
- Have $P(A_i)$, for every i initial "beliefs"
- Have $P(B | A_i)$, for every i

revised "beliefs," given that B occurred:

$$P(A_i | B) = \frac{P(A_i \cap B)}{P(B)}$$

$$P(B | A_i) = \frac{P(A_i \cap B)}{P(A_i)}$$

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{P(B)}$$

$$P(A_i | B) = \frac{P(A_i) P(B | A_i)}{\sum_j P(A_j) P(B | A_j)}$$

Bayes' rule and inference

- Thomas Bayes, presbyterian minister (c. 1701-1761)
- “Bayes’ theorem,” published posthumously
- systematic approach for incorporating new evidence
- Bayesian inference
 - initial beliefs $P(A_i)$ on possible causes of an observed event B
 - model of the world under each A_i : $P(B | A_i)$

$$A_i \xrightarrow[\mathbf{P}(B | A_i)]{\text{model}} B$$

- draw conclusions about causes

$$B \xrightarrow[\mathbf{P}(A_i | B)]{\text{inference}} A_i$$

Regras são consequências dos axiomas e da definição de prob. condicional.

Permitem trabalhar com mundos hipotéticos: se eu soubesse que B aconteceu eu saberia continuar resolvendo o meu problema. Nesse caso, condiciona em B e usa os resultados.

$$P(A|B) = P(B|A) \text{ se } P(A) = P(B)$$

Kay: 3.16, 3.15, 3.23, 3.26

Kay: 4.1, 4.2, 4.7, 4.9, 4.10, 4.11, 4.16, 4.19, 4.23, ~~4.26~~, 4.34, 4.39

$$P(A|B) = P(B|A) \frac{P(A)}{P(B)}$$

Mostre que a probabilidade, definida como a frequência relativa, satisfaz os axiomas de Kolmogorov

<https://betterexplained.com/articles/an-intuitive-and-short-explanation-of-bayes-theorem/>

1% of women have breast cancer (and therefore 99% do not).

80% of mammograms detect breast cancer when it is there (and therefore 20% miss it).

9.6% of mammograms detect breast cancer when it's not there (and therefore 90.4% correctly return a negative result).

If you get a positive result, what is the probability that you actually have cancer?

990		10	
S+	95	8	C+
S-	895	2	C-

B: 103 mulheres

$$P(C|+) = \frac{8}{103} \approx 8\%$$

Encontre um exemplo que mostre que $P(X|A) + P(X|A^c) \neq 1$. O ponto é o seguinte: quando dizemos que a probabilidade condicional é uma probabilidade normal, assumimos que o evento condicionante é fixo. O fato é que um modelo probabilístico num mundo em que o evento A é verdade é diferente do modelo em que A é falso, ou seja, A^c é verdade. A afirmação correta é dizer que $P(X^c|A) + P(X|A) = 1$, pois ambos estão condicionados ao mesmo evento A .

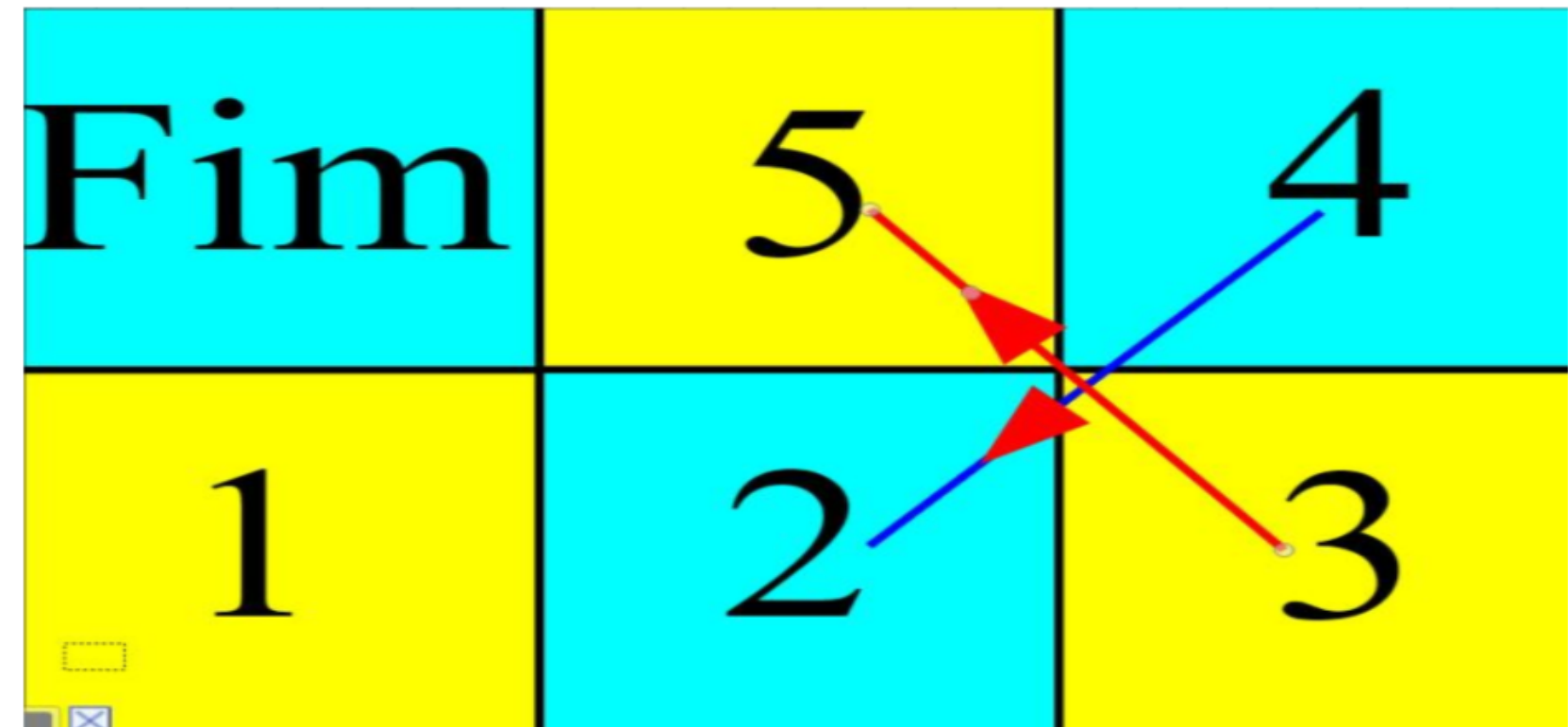
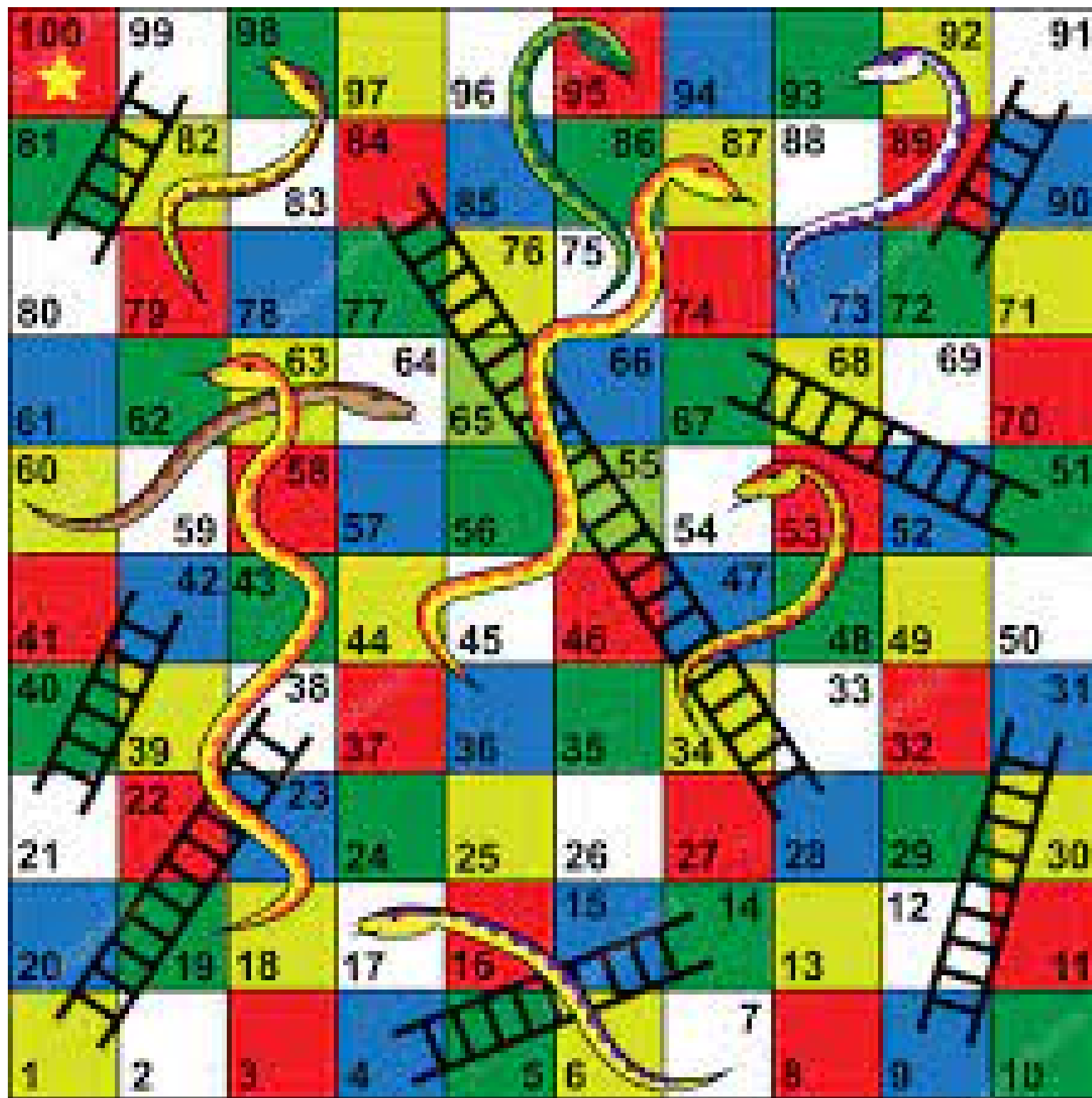
Esboce um diagrama de Venn de dois eventos tais que $P(B|A) = P(B)$. Pode associar a área de cada conjunto do diagrama à probabilidade do evento.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B,C) = \frac{P(A \cap B|C)}{P(B|C)}$$

$$P(A|C) = \frac{P(A \cap C)}{P(C)} \quad \longrightarrow$$

$$= \frac{P(A \cap C|B)}{P(C|B)}$$



<https://www.datagenetics.com/blog/november12011/>

Dado é uma moeda, anda uma ou duas casas.
 Calcule a prob de estar na casa 5 na rodada 2.

$$P(1, R_1) = \frac{1}{2} \quad P(3, R_1) = 0$$

$$P(2, R_1) = \frac{1}{2} \quad P(4, R_1) = 0$$